# Communication Tone, Pull Request Outcomes, and Contributor Return in Open Source Software
# Part 1: Initial Presentation

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# Part 1
# Study Objective and Research Questions

## Objective

Build a reproducible pipeline linking PR discussion tone to:

- merge outcome
- contributor return (30 / 90 / 180 days)

while controlling for contributor history and discussion structure.

## Research Questions

- RQ1: Communication signals vs **merge**
- RQ2: Communication signals vs **return**
- RQ3: **Newcomers vs experienced** contributors
- RQ4: Role of **politeness**

## Part 2
# Why this question is important

### Why it matters

- Open source coordination happens through **text-based PR discussion**
- Discussion is part of the **contribution process itself**
- Directly linked to:
  - contribution success
  - contributor retention

### Impact

Understanding these signals helps evaluate **community health** and review practices.

# Part 2
# Why it is interesting

### Research value

- Combines **repository mining + NLP**
- Uses real-world OSS data
- Tests common assumptions empirically

### Why I chose it

- PR discussions are central to OSS
- Socially meaningful + measurable
- Full pipeline:
  - data collection
  - feature design
  - modeling

**Result matters either way:**

Tone matters $\Rightarrow$ insight / Tone doesn't $\Rightarrow$ challenge assumptions

# Part 3
## Data: Repositories

| Repository | Domain | Language |
|---|---|---|
| kubernetes/kubernetes | Cloud infrastructure | Go |
| vercel/next.js | Web framework | JS / TS |
| scikit-learn/scikit-learn | Machine learning | Python |

- Large PR volume
- Different domains and communities
- Active review culture

# Part 3
# Unit of Analysis

### One row = one PR

The focal contributor is the PR author.

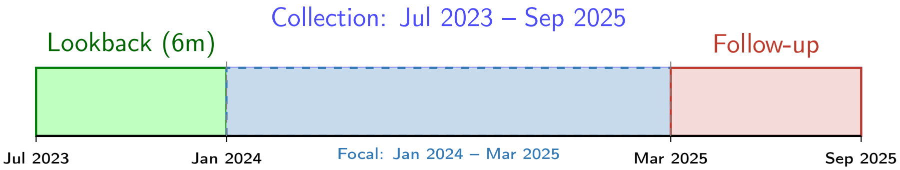
**Feature layers**

- **Outcome**: merged, returned
- **Structural**: comments, participants, delay, history
- **Communication**: toxicity, sentiment, anger, politeness

### Key design

Only **received comments** are used for NLP features.

# Part 3
# Time Windows




- **Focal PRs**: Jan 2024 – Mar 2025 (analysis set)
- **Lookback**: classify newcomers
- **Follow-up**: measure return

**Why this design**

Avoids **mislabeling experience** and **incomplete return data**

# Part 3
# Methods and Pipeline

## Data collection

GitHub REST API:

- PR metadata
- issue comments
- review comments

## NLP signals

- toxicity (max + thresholds)
- sentiment
- anger
- politeness

# Part 3
## Challenges

- Toxicity likely sparse
- Technical language may confuse NLP
- Experience may dominate outcomes
- Observational study (no causality)
- PR-only return misses other activity

**Expectation**

Communication tone may still matter through **sentiment, politeness, and interaction structure**

Thank you